Causality and dealing with omitted variables bias

Econometrics. Lecture 7

The CLRM assumptions. Review

Assumption MLR.1 (Linear in parameters)	$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$
Assumption MLR.2 (Random sampling)	A random sample of *n* observations, $\{(x_{i1}, x_{i2}, \ldots, x_{ik}, y_i): i = 1, 2, \ldots, n\}$
Assumption MLR.3 (No perfect collinearity)	None of the independent variables is constant, and there are no exact linear relationships among them
Assumption MLR.4 (Zero conditional mean)	$\mathrm{E}(u \mid x_1, x_2, \ldots, x_k) = 0$
Assumption MLR.5 (Homoskedasticity)	$\mathrm{Var}(u \mid x_1, x_2, \ldots, x_k) = \sigma^2$
Assumption MLR.6 (Normality)	$u \sim \mathrm{Normal}(0, \sigma^2)$.

Content

1. Functional form misspecification
2. Using proxy variables for unobserved explanatory variables
3. Instrumental variables and two stage least squares
4. Policy analysis with pooled cross sections

1. Functional form misspecification

Functional form misspecification

Suppose, that an hourly wage is determined by

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + \beta_4 female + \beta_5 female \cdot educ + u,$$

When we don't use logs for a dependent variable or omit an interaction term, we are committing a functional form misspecification.

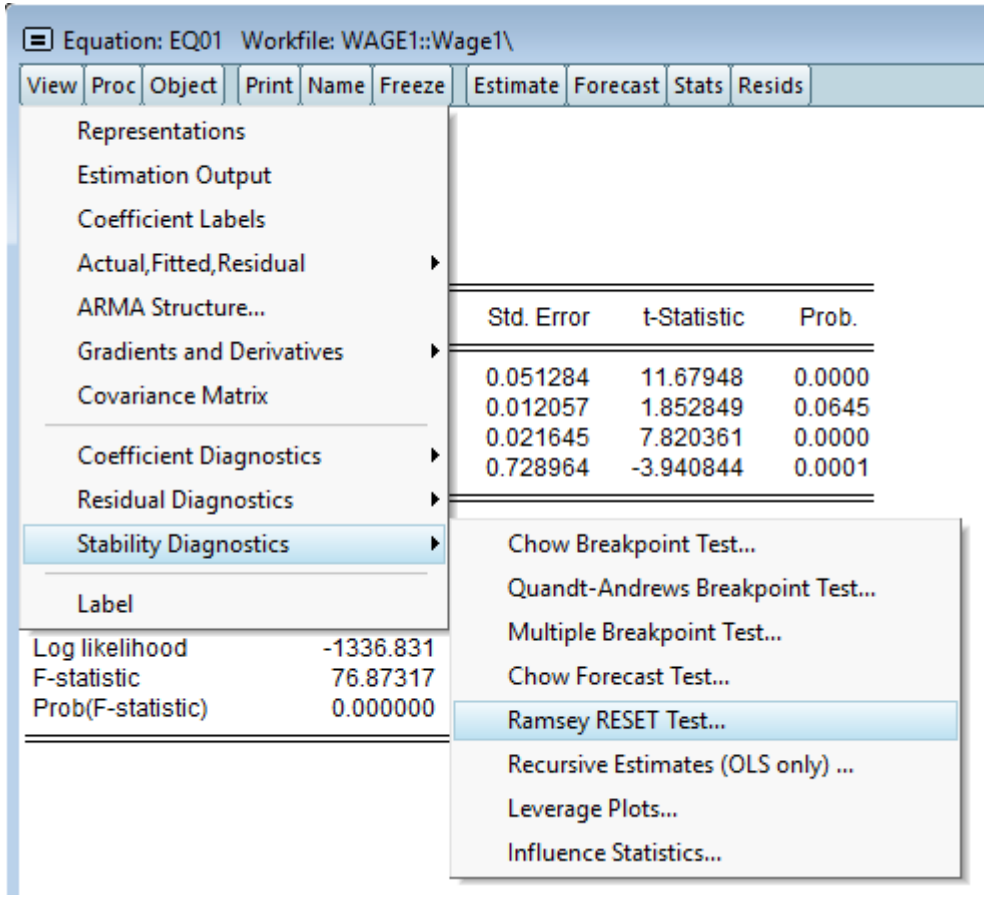
Misspecifying the functional form of a model generally leads to biased and inconsistent estimators of β_j.

Detecting functional form misspecification

It often makes sense to add quadratic terms of any significant variables to a model and to perform a **joint test of significance**. If the additional quadratics are significant, they can be added to the model (at the cost of complicating the interpretation of the model).

In many cases, **using logarithms** of certain variables and **adding quadratics** are sufficient for detecting many important nonlinear relationships in economics.

Test for misspecification / omitted variables



Reject H_0: no functional-form misspecification / no omitted nonlinear terms

Ramsey RESET Test
Equation: EQ01
Omitted Variables: Squares of fitted values
Specification: WAGE EDUC EXPER TENURE C

	Value	df	Probability
t-statistic	4.565659	521	0.0000
F-statistic	20.84525	(1, 521)	0.0000
Likelihood ratio	20.63519	1	0.0000

F-test summary:

	Sum of Sq.	df	Mean Squares
Test SSR	191.0579	1	191.0579
Restricted SSR	4966.303	522	9.513990
Unrestricted SSR	4775.245	521	9.165537

LR test summary:

	Value
Restricted LogL	-1336.831
Unrestricted LogL	-1326.513

Unrestricted Test Equation:
Dependent Variable: WAGE
Method: Least Squares
Date: 12/28/25 Time: 22:13
Sample: 1 526
Included observations: 526

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EDUC	-0.014094	0.143401	-0.098284	0.9217
EXPER	8.38E-05	0.012799	0.006549	0.9948
TENURE	-0.057103	0.053941	-1.058611	0.2903
C	2.823106	1.438152	1.963009	0.0502
FITTED^2	0.090923	0.019915	4.565659	0.0000

R-squared	0.333105	Mean dependent var	5.896103
Adjusted R-squared	0.327985	S.D. dependent var	3.693086
S.E. of regression	3.027464	Akaike info criterion	5.062788
Sum squared resid	4775.245	Schwarz criterion	5.103332
Log likelihood	-1326.513	Hannan-Quinn criter.	5.078663
F-statistic	65.05809	Durbin-Watson stat	1.801762
Prob(F-statistic)	0.000000		

2. Using proxy variables for unobserved explanatory variables

Omitted variable bias

Population model $\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 ability + u$

If, education is correlated with ability, then putting ability in the error term causes the OLS estimator of β_1 (and β_2) to be biased.

One possibility to solve, or at least mitigate, the omitted variables bias is to obtain a proxy variable for the omitted variable.

A **proxy variable** is something that is related to the unobserved variable that we would like to control for in our analysis.

In the wage equation, one possibility is to use IQ as a proxy for ability.

Plug-in solution to the omitted variables problem

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3^* + u$$

Data are available on y, x_1, and x_2. The explanatory variable x_3^* is unobserved.

x_3 is an observed proxy variable for x_3^*. x_3 should have some relationship to x_3^*.

The plug-in solution to the omitted variables problem

Run the regression of y on x_1, x_2, x_3.

The assumptions needed for the plug-in solution to provide consistent estimators of β_1 and β_2 are:

1) *u* is uncorrelated with x_3;

2) once x_3 is controlled for, the expected value of x_3^* does not depend on x_1 or x_2.

IQ as a proxy for ability

Model 1

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 01/08/26 Time: 12:53
Sample: 1 935
Included observations: 935

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EDUC	0.065431	0.006250	10.46826	0.0000
EXPER	0.014043	0.003185	4.408852	0.0000
TENURE	0.011747	0.002453	4.788998	0.0000
MARRIED	0.199417	0.039050	5.106691	0.0000
SOUTH	-0.090904	0.026249	-3.463193	0.0006
URBAN	0.183912	0.026958	6.822087	0.0000
BLACK	-0.188350	0.037667	-5.000444	0.0000
C	5.395497	0.113225	47.65286	0.0000

R-squared	0.252558	Mean dependent var	6.779004
Adjusted R-squared	0.246914	S.D. dependent var	0.421144
S.E. of regression	0.365471	Akaike info criterion	0.833260
Sum squared resid	123.8185	Schwarz criterion	0.874676
Log likelihood	-381.5490	Hannan-Quinn criter.	0.849052
F-statistic	44.74706	Durbin-Watson stat	1.822637
Prob(F-statistic)	0.000000		

Model 2

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 01/08/26 Time: 12:52
Sample: 1 935
Included observations: 935

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EDUC	0.054411	0.006928	7.853173	0.0000
EXPER	0.014146	0.003165	4.469316	0.0000
TENURE	0.011395	0.002439	4.671302	0.0000
MARRIED	0.199764	0.038802	5.148237	0.0000
SOUTH	-0.080169	0.026253	-3.053735	0.0023
URBAN	0.181946	0.026793	6.790848	0.0000
BLACK	-0.143125	0.039492	-3.624118	0.0003
IQ	0.003559	0.000992	3.588501	0.0004
C	5.176439	0.128001	40.44074	0.0000

R-squared	0.262809	Mean dependent var	6.779004
Adjusted R-squared	0.256441	S.D. dependent var	0.421144
S.E. of regression	0.363152	Akaike info criterion	0.821588
Sum squared resid	122.1203	Schwarz criterion	0.868182
Log likelihood	-375.0925	Hannan-Quinn criter.	0.839355
F-statistic	41.26501	Durbin-Watson stat	1.820310
Prob(F-statistic)	0.000000		

Using lagged dependent variables as proxies

Using a **lagged dependent variable** in a cross-sectional equation increases the data requirements, but it also provides a simple way to account for historical factors that cause current differences in the dependent variable that are difficult to account for in other ways.

A simple equation to explain city crime rates

$$crime = \beta_0 + \beta_1 unem + \beta_2 expend + \beta_3 crime_{-1} + u$$

where *crime* is a measure of per capita crime, *unem* is the city unemployment rate, *expend* is per capita spending on law enforcement, and $crime_{-1}$ indicates the crime rate measured in some earlier year.

The main reason for putting $crime_{-1}$ in the equation is that cities with high historical crime rates may spend more on crime prevention. Thus, unobserved factors that affect crime are likely to be correlated with *expend* (and *unem*).

City crime rates

No lagged dependent variable

Dependent Variable: LOG(CRMRTE)
Method: Least Squares
Date: 09/04/26 Time: 09:09
Sample: 1 92 IF YEAR=87
Included observations: 46

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	3.342899	1.250526	2.673194	0.0106
UNEM	-0.029003	0.032339	-0.896856	0.3748
LOG(LAWEXPC)	0.203365	0.172653	1.177881	0.2453

R-squared	0.057117	Mean dependent var	4.590739
Adjusted R-squared	0.013262	S.D. dependent var	0.325302
S.E. of regression	0.323138	Akaike info criterion	0.641519
Sum squared resid	4.489982	Schwarz criterion	0.760778
Log likelihood	-11.75494	Hannan-Quinn criter.	0.686194
F-statistic	1.302394	Prob(F-statistic)	0.282389

With lagged dependent variable

Dependent Variable: LOG(CRMRTE)
Method: Least Squares
Date: 09/04/26 Time: 09:12
Sample: 1 92 IF YEAR=87
Included observations: 46

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.076451	0.821143	0.093103	0.9263
UNEM	0.008621	0.019517	0.441726	0.6610
LOG(LAWEXPC)	-0.139576	0.108641	-1.284745	0.2059
LCRMRT_1	1.193922	0.132099	9.038121	0.0000

R-squared	0.679830	Mean dependent var	4.590739
Adjusted R-squared	0.656960	S.D. dependent var	0.325302
S.E. of regression	0.190528	Akaike info criterion	-0.395092
Sum squared resid	1.524641	Schwarz criterion	-0.236080
Log likelihood	13.08712	Hannan-Quinn criter.	-0.335525
F-statistic	29.72673	Prob(F-statistic)	0.000000

3. Instrumental variables and two stage least squares

Motivation

$$y = \beta_0 + \beta_1 x + u$$

We think that *x* and *u* are correlated:

$$\mathrm{Cov}(x, u) \neq 0$$

Let *z* be an observable variable that is uncorrelated with *u* but correlated with *x*.

Instrument exogeneity

$$\mathrm{Cov}(z, u) = 0$$

Instrument relevance

$$\mathrm{Cov}(z, x) \neq 0$$

Then, we call *z* an **instrumental variable (IV)** for *x*, or sometimes simply an **instrument** for *x*.

Instruments. Example

Is IQ a good candidate as an instrumental variable for education?

$$\log(wage) = \beta_0 + \beta_1 educ + u$$

IQ is highly correlated with ability – a **good proxy**.

IQ is also correlated with u – a **poor instrument** .

Labor economists have used family background variables as IVs for education.

Another IV choice for education in is number of siblings while growing up.

IV estimation of the multiple regression model (1)

$$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + u_1 \qquad \log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + u$$

The variables y_2 $(educ)$ and z_1 $(exper)$ are the explanatory variables, and u_1 is the error.

We use z_1 $(exper)$ to indicate that this variable is **exogenous**.

We use y_2 $(educ)$ to indicate that this variable is suspected of being correlated with u_1. y_2 $(educ)$ is **endogenous explanatory variable**.

If the equation is estimated by OLS, all the estimators will be biased and inconsistent.

We need another exogenous variable – z_2 – that does not appear in the equation as an instrument for y_2.

IV estimation of the multiple regression model (2)

The instrumental variable z_2 must be correlated with y_2. But because of the presence of z_1 we now need to state the assumption in terms of *partial correlation*.

$$y_2 = \pi_0 + \pi_1 z_1 + \pi_2 z_2 + v_2$$

$$\pi_2 \neq 0$$

After partialling out z_1, y_2 and z_2 must still be correlated.

Two stage least squares

$$\log(wage) = \beta_0 + \beta_1 educ + u$$

OLS uses all variation in education. If education is endogenous, part of its variation is **correlated with the error term** (ability, motivation, etc.).

$$educ = exogenous\ variation + endogenous\ variation$$

To obtain unbiased and consistent estimators we should eliminate the endogenous variation.

Two stage least squares.
First stage

We use instrument (father's education '*fatheduc*') to isolate the exogenous variation in the endogenous explanatory variable (education).

$$\widehat{educ} = \pi_0 + \pi_1 fatheduc$$

Fitted values ($\widehat{educ}$) is the part of education that:

- is correlated with education (relevance);
- is uncorrelated with the wage error term u (exogeneity).

valid for estimating the causal effect of education on wage

$$educ = \underbrace{\widehat{educ}}_{exogenous} + \underbrace{(educ - \widehat{educ})}_{endogenous}$$

affects wages directly through ability, motivation, etc. → invalid for causality

Two stage least squares (2SLS). Second stage

We use fitted values ($\widehat{educ}$) as an instrument for $educ$ in the wage equation

$$\log(\widehat{wage}) = \hat{\beta}_0 + \hat{\beta}_1 \widehat{educ}$$

In most cases you should avoid doing the second stage manually, as the standard errors and test statistics obtained in this way are not valid.

Any regression software that supports 2SLS asks for the dependent variable, the list of explanatory variables (both exogenous and endogenous), and the entire list of instrumental variables (that is, all exogenous variables). The output is typically quite similar to that for OLS.

Return to education for married women (1)

We use the data on married working women to estimate the return to education in the simple regression model.

$$\log(wage) = \beta_0 + \beta_1 educ + u$$

OLS

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 01/05/26 Time: 16:57
Sample (adjusted): 1 428
Included observations: 428 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.185197	0.185226	-0.999843	0.3180
EDUC	0.108649	0.014400	7.545125	0.0000

R-squared	0.117883	Mean dependent var	1.190173
Adjusted R-squared	0.115812	S.D. dependent var	0.723198
S.E. of regression	0.680032	Akaike info criterion	2.071309
Sum squared resid	197.0010	Schwarz criterion	2.090276
Log likelihood	-441.2600	Hannan-Quinn criter.	2.078800
F-statistic	56.92891	Durbin-Watson stat	1.984707
Prob(F-statistic)	0.000000		

Educ and *fatheduc*

Dependent Variable: EDUC
Method: Least Squares
Date: 01/08/26 Time: 15:23
Sample: 1 428
Included observations: 428

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	10.23705	0.275936	37.09933	0.0000
FATHEDUC	0.269442	0.028586	9.425538	0.0000

R-squared	0.172560	Mean dependent var	12.65888
Adjusted R-squared	0.170617	S.D. dependent var	2.285376
S.E. of regression	2.081302	Akaike info criterion	4.308526
Sum squared resid	1845.354	Schwarz criterion	4.327494
Log likelihood	-920.0246	Hannan-Quinn criter.	4.316017
F-statistic	88.84076	Durbin-Watson stat	1.918691
Prob(F-statistic)	0.000000		

IV

Dependent Variable: LOG(WAGE)
Method: Two-Stage Least Squares
Date: 01/05/26 Time: 16:58
Sample (adjusted): 1 428
Included observations: 428 after adjustments
Instrument specification: FATHEDUC
Constant added to instrument list

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.441103	0.446102	0.988795	0.3233
EDUC	0.059173	0.035142	1.683850	0.0929

R-squared	0.093438	Mean dependent var	1.190173
Adjusted R-squared	0.091310	S.D. dependent var	0.723198
S.E. of regression	0.689390	Sum squared resid	202.4601
F-statistic	2.835351	Durbin-Watson stat	1.968194
Prob(F-statistic)	0.092943	Second-Stage SSR	221.9799
J-statistic	2.83E-43	Instrument rank	2

We use father's education (*fatheduc*) as an instrumental variable for *educ*.

Return to education for married women (2)

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + u$$

Step 1

We use father's education (*fatheduc*) and mother's education (*motheduc*) as an instrumental variable for *educ*.

Dependent Variable: EDUC
Method: Least Squares
Date: 01/08/26 Time: 19:03
Sample: 1 428
Included observations: 428

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	9.102640	0.426561	21.33958	0.0000
EXPER	0.045225	0.040251	1.123593	0.2618
EXPER^2	-0.001009	0.001203	-0.838572	0.4022
FATHEDUC	0.189548	0.033756	5.615173	0.0000
MOTHEDUC	0.157597	0.035894	4.390609	0.0000

R-squared	0.211471	Mean dependent var	12.65888
Adjusted R-squared	0.204014	S.D. dependent var	2.285376
S.E. of regression	2.038967	Akaike info criterion	4.274378
Sum squared resid	1758.575	Schwarz criterion	4.321797
Log likelihood	-909.7168	Hannan-Quinn criter.	4.293106
F-statistic	28.36041	Durbin-Watson stat	1.939888
Prob(F-statistic)	0.000000		

Education is (partially) correlated with parents' education.

Redundant Variables Test
Equation: EQ03_0
Redundant variables: FATHEDUC MOTHEDUC
Specification: EDUC C EXPER EXPER^2 FATHEDUC MOTHEDUC
Null hypothesis: FATHEDUC MOTHEDUC are jointly insignificant

	Value	df	Probability
F-statistic	55.40030	(2, 423)	0.0000
Likelihood ratio	99.57428	2	0.0000

Return to education for married women (3)

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + u$$

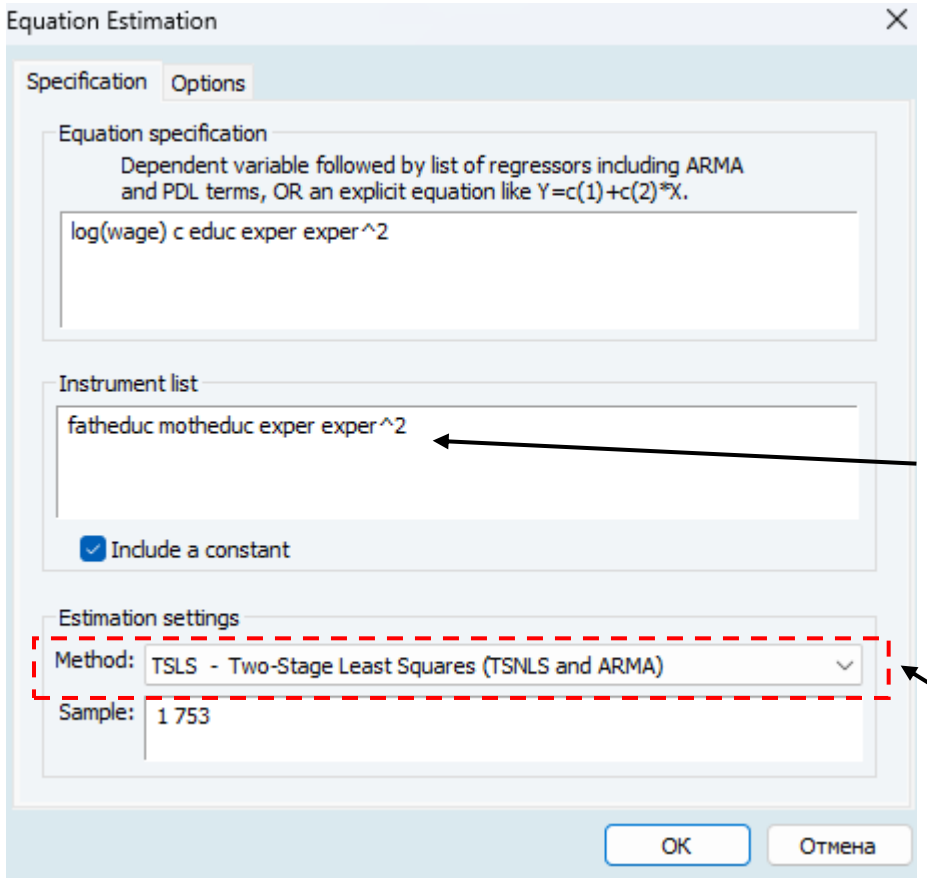
Step 2

Estimation of the equation by 2SLS.

Dependent Variable: LOG(WAGE)
Method: Two-Stage Least Squares
Date: 01/08/26 Time: 19:16
Sample (adjusted): 1 428
Included observations: 428 after adjustments
Instrument specification: FATHEDUC MOTHEDUC EXPER EXPER^2
Constant added to instrument list

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.048100	0.400328	0.120152	0.9044
EDUC	0.061397	0.031437	1.953024	0.0515
EXPER	0.044170	0.013432	3.288329	0.0011
EXPER^2	-0.000899	0.000402	-2.237993	0.0257

R-squared	0.135708	Mean dependent var	1.190173
Adjusted R-squared	0.129593	S.D. dependent var	0.723198
S.E. of regression	0.674712	Sum squared resid	193.0200
F-statistic	8.140709	Durbin-Watson stat	1.945659
Prob(F-statistic)	0.000028	Second-Stage SSR	212.2096
J-statistic	0.374538	Instrument rank	5
Prob(J-statistic)	0.540541		



Instrument list must include all exogenous explanatory variables and as many excluded exogenous variables as there are included endogenous explanatory variables

Select TSLS as method

Do not reject the null hypothesis that instruments are exogenous

Multiple endogenous explanatory variables

Two stage least squares can also be used in models with more than one endogenous explanatory variable.

Order condition for Identification of an equation:

- we need at least as many excluded exogenous variables (instruments) as there are included endogenous explanatory variables in the structural equation.

2SLS with heteroskedasticity

Test for heteroskedasticity

Heteroskedasticity Test: Breusch-Pagan-Godfrey
Null hypothesis: Homoskedasticity

F-statistic	4.074511	Prob. F(3,424)	0.0072
Obs*R-squared	11.99310	Prob. Chi-Square(3)	0.0074
Scaled explained SS	32.27196	Prob. Chi-Square(3)	0.0000

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 01/08/26 Time: 19:30
Sample: 1 428
Included observations: 428

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.600496	0.311793	1.925943	0.0548
EDUC	0.023186	0.022206	1.044132	0.2970
EXPER	-0.055671	0.020681	-2.691858	0.0074
EXPER^2	0.001205	0.000617	1.951736	0.0516

R-squared	0.028021	Mean dependent var	0.450981
Adjusted R-squared	0.021144	S.D. dependent var	1.057320
S.E. of regression	1.046082	Akaike info criterion	2.937283
Sum squared resid	463.9780	Schwarz criterion	2.975218
Log likelihood	-624.5785	Hannan-Quinn criter.	2.952265
F-statistic	4.074511	Durbin-Watson stat	1.613138
Prob(F-statistic)	0.007164		

Robust standard errors

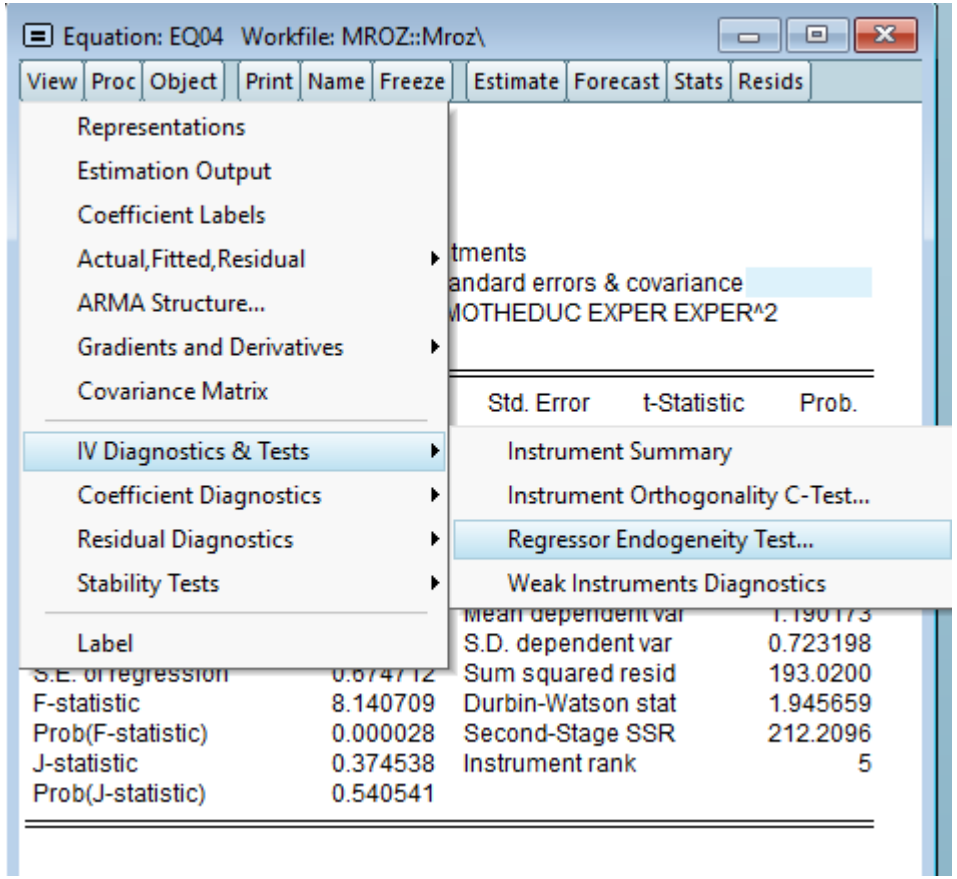
Dependent Variable: LOG(WAGE)
Method: Two-Stage Least Squares
Date: 01/08/26 Time: 19:31
Sample (adjusted): 1 428
Included observations: 428 after adjustments
White heteroskedasticity-consistent standard errors & covariance
Instrument specification: FATHEDUC MOTHEDUC EXPER EXPER^2
Constant added to instrument list

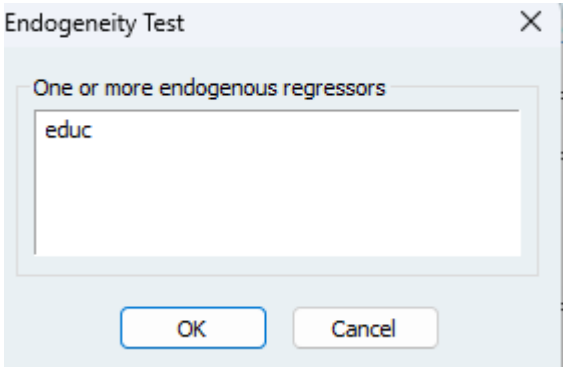
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.048100	0.429798	0.111914	0.9109
EDUC	0.061397	0.033339	1.841609	0.0662
EXPER	0.044170	0.015546	2.841202	0.0047
EXPER^2	-0.000899	0.000430	-2.090220	0.0372

R-squared	0.135708	Mean dependent var	1.190173
Adjusted R-squared	0.129593	S.D. dependent var	0.723198
S.E. of regression	0.674712	Sum squared resid	193.0200
F-statistic	8.140709	Durbin-Watson stat	1.945659
Prob(F-statistic)	0.000028	Second-Stage SSR	212.2096
J-statistic	0.374538	Instrument rank	5
Prob(J-statistic)	0.540541		

Testing for endogeneity

Hausman (1978) suggested directly comparing the OLS and 2SLS estimates and determining whether the differences are statistically significant.





Endogeneity Test
Equation: EQ04
Endogenous variables to treat as exogenous: EDUC
Specification: LOG(WAGE) C EDUC EXPER EXPER^2
Instrument specification: C FATHEDUC MOTHEDUC EXPER EXPER^2
Null hypothesis: EDUC are exogenous

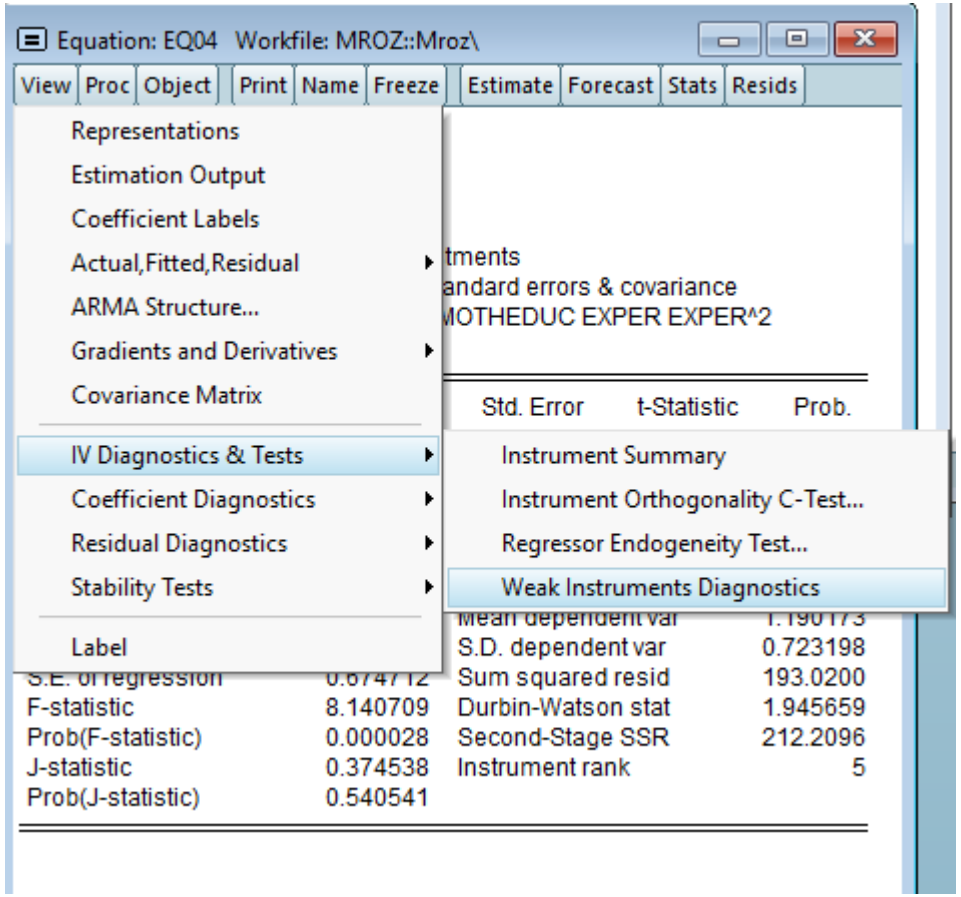
	Value	df	Probability
Difference in J-stats	2.780835	1	0.0954

J-statistic summary:

	Value
Restricted J-statistic	3.164751
Unrestricted J-statistic	0.383916

Weak instruments diagnostics

Problem of **weak instruments** – "low" (but not zero) correlation between *z* and *x*.



The null hypothesis of weak instruments is rejected if Cragg–Donald F-statistic > Stock–Yogo critical values

Weak Instrument Diagnostics
Equation: EQ04

Cragg-Donald F-stat:	55.40030

Stock-Yogo bias critical values not available for models with less than 3 instruments.

Stock-Yogo critical values (size):

10%	19.93
15%	11.59
20%	8.75
25%	7.25

Moment selection criteria:

SIC-based:	-5.684585
HQIC-based:	-3.246608
Relevant MSC:	-15.98390

Criticisms of IV estimation

- Questionable validity of instruments (exogeneity). What if *fatheduc* relates to female ability? We cannot test this…
- Weak instruments. Even valid instruments may be **weakly correlated** with endogenous regressors. Strong, credible instruments are rare.
- Loss of efficiency (large standard errors). IV uses only instrument-induced variation. Results may be statistically insignificant even when effects are economically meaningful.
- Sensitivity to specification and instrument choice.

4. Policy analysis with pooled cross sections

Pooled cross section. Review

If a random sample is drawn at each time period, pooling the resulting random samples gives us an **independently pooled cross section.**

For instance, in each year, we can draw a random sample on hourly wages, education, experience, and so on from the population of working people in Lithuania.

By pooling random samples drawn from the same population, but at different points in time, we can get more precise estimators and test statistics with more power.

Pooled cross section. Estimation

To reflect the fact that the population may have different distributions in different time periods, we allow the intercept to differ across periods, usually years. This is easily accomplished by including year dummy variables

A log(*wage*) equation pooled across the years 1978 (the base year) and 1985 is

$$\log(wage) = \beta_0 + \delta_0 y85 + \beta_1 educ + \delta_1 y85 * educ + \beta_2 exper + \beta_3 exper^2 + \\ + \beta_4 union + \beta_5 female + \delta_2 y85 * female + u$$

where *union* is a dummy variable equal to one if the person belongs to a union, and zero otherwise; *y85* is a dummy variable equal to one if the observation comes from 1985 and zero if it comes from 1978.

There are 550 people in the sample in 1978 and a different set of 534 people in 1985.

Changes in the return to education and the gender wage gap

We use the data to estimate the equation

Dependent Variable: LWAGE
Method: Least Squares
Date: 01/09/26 Time: 12:53
Sample: 1 1084
Included observations: 1084

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.458933	0.093449	4.911078	0.0000
Y85	0.117806	0.123782	0.951725	0.3415
EDUC	0.074721	0.006676	11.19174	0.0000
Y85*EDUC	0.018461	0.009354	1.973509	0.0487
EXPER	0.029584	0.003567	8.293165	0.0000
EXPER^2	-0.000399	7.75E-05	-5.151307	0.0000
UNION	0.202132	0.030294	6.672233	0.0000
FEMALE	-0.316709	0.036621	-8.648173	0.0000
Y85*FEMALE	0.085052	0.051309	1.657644	0.0977

R-squared	0.426186	Mean dependent var	1.867301
Adjusted R-squared	0.421915	S.D. dependent var	0.542804
S.E. of regression	0.412704	Akaike info criterion	1.076097
Sum squared resid	183.0991	Schwarz criterion	1.117513
Log likelihood	-574.2443	Hannan-Quinn criter.	1.091776
F-statistic	99.80353	Durbin-Watson stat	1.918367
Prob(F-statistic)	0.000000		

Hourly wage here is in nominal (or current) dollars. Should we transform the wage into real terms in our model?

Causality

Economists are often interested in policy analysis.

Does job training improve chances of becoming employed?

Will the construction of incinerator affect house prices?

Other examples?

Effect of worker compensation laws on weeks out of work

Meyer, Viscusi, and Durbin (1995) (hereafter, MVD) studied the length of time (in weeks) that an injured worker receives workers' compensation.

On July 15, 1980, Kentucky raised the cap on weekly earnings that were covered by workers' compensation.

The hypothesis is that an increase in the cap has no effect on the benefit for low-income workers, but it makes it less costly for a high-income worker to stay on workers' compensation.

Naïve analysis

Let *durat* denotes the length of time (in weeks) that an injured worker receives workers' compensation.

A naïve analyst would use only the data after the policy change and estimate a very simple model

Dependent Variable: LOG(DURAT)
Method: Least Squares
Date: 01/09/26 Time: 15:51
Sample: 1 7150 IF KY=1 AND AFCHNGE=1
Included observations: 2688

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.133273	0.032858	34.48971	0.0000
HIGHEARN	0.447080	0.049997	8.942142	0.0000

R-squared	0.028909	Mean dependent var	1.326375
Adjusted R-squared	0.028548	S.D. dependent var	1.302728
S.E. of regression	1.283998	Akaike info criterion	3.338579
Sum squared resid	4428.278	Schwarz criterion	3.342966
Log likelihood	-4485.050	Hannan-Quinn criter.	3.340165
F-statistic	79.96190	Durbin-Watson stat	1.917876
Prob(F-statistic)	0.000000		

where *highearn* is a binary variable equal to one if the worker is high-income, and zero otherwise.

Question: Why is this naïve approach?

The same regression for a period before the policy change

However, the data for a period before the policy change show:

Dependent Variable: LOG(DURAT)
Method: Least Squares
Date: 01/09/26 Time: 15:51
Sample: 1 7150 IF KY=1 AND AFCHNGE=0
Included observations: 2938

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.125615	0.030405	37.02113	0.0000
HIGHEARN	0.256479	0.046934	5.464700	0.0000

R-squared	0.010069	Mean dependent var	1.233253
Adjusted R-squared	0.009732	S.D. dependent var	1.261613
S.E. of regression	1.255459	Akaike info criterion	3.293560
Sum squared resid	4627.656	Schwarz criterion	3.297635
Log likelihood	-4836.240	Hannan-Quinn criter.	3.295027
F-statistic	29.86295	Durbin-Watson stat	1.892153
Prob(F-statistic)	0.000000		

the average duration of benefits for high-income workers was 26% higher than the average duration for low-income workers.

Thus, high-income workers have higher duration of benefits even before the policy change.

Question: How, then, can we tell whether the policy change influence on the duration of benefits for high-income workers?

Difference-in-differences

The key is to look at how the coefficient on *highearn* changed between two periods.

The difference in the two coefficients on *highearn* is

$$\hat{\delta}_1 = 0.447080 - 0.256479 = 0.190601 \approx 19 \text{ p. p.}$$

$\hat{\delta}_1$ has become known as the **difference-in-differences estimator** because it can be expressed as

$$\hat{\delta}_1 = (\overline{\log(durat)}_{a,hi} - \overline{\log(durat)}_{a,li}) - (\overline{\log(durat)}_{b,hi} - \overline{\log(durat)}_{b,li})$$

where hi stands for "high-income workers" and li stands for "low-income workers"; a stands for a period after the policy measure and b – before.

Difference-in-differences. Interpretation

To test whether $\hat{\delta}_1$ is statistically different from zero, we need to find its standard error by using a regression analysis based on data pooled over both periods

$$\log(durat) = \beta_0 + \delta_0 afchnge + \beta_1 highearn + \delta_1 afchnge * highearn + u$$

β_0 is the average duration of benefits for **low-income workers before** the policy change.

δ_0 captures changes in the duration of benefits for **low-income workers after** the change. ($afchnge$ is a binary variable equal to one for the period after the measure)

β_1 measures the changes in the duration **between two groups** of workers **not** because of the policy change.

δ_1 measures the changes in the duration **between two groups** of workers **due to** the policy change (everything else fixed).

Difference-in-differences. Estimation

Simple model

Dependent Variable: LOG(DURAT)
Method: Least Squares
Date: 01/09/26 Time: 15:39
Sample: 1 7150 IF KY=1
Included observations: 5626

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.125615	0.030737	36.62107	0.0000
AFCHNGE	0.007657	0.044717	0.171238	0.8640
HIGHEARN	0.256479	0.047446	5.405646	0.0000
AFCHNGE*HIGHEARN	0.190601	0.068509	2.782138	0.0054

R-squared	0.020663	Mean dependent var	1.277745
Adjusted R-squared	0.020140	S.D. dependent var	1.282151
S.E. of regression	1.269174	Akaike info criterion	3.315321
Sum squared resid	9055.934	Schwarz criterion	3.320038
Log likelihood	-9321.997	Hannan-Quinn criter.	3.316964
F-statistic	39.53958	Durbin-Watson stat	1.904874
Prob(F-statistic)	0.000000		

Extended model

Dependent Variable: LOG(DURAT)
Method: Least Squares
Date: 01/09/26 Time: 16:34
Sample: 1 7150 IF KY=1
Included observations: 5349

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.907320	0.049495	18.33138	0.0000
AFCHNGE	0.045169	0.041456	1.089565	0.2760
HIGHEARN	0.139914	0.047757	2.929708	0.0034
AFCHNGE*HIGHEARN	0.178105	0.064164	2.775758	0.0055
MALE	-0.102215	0.041046	-2.490253	0.0128
MARRIED	0.115256	0.036097	3.192928	0.0014
CONSTRUC	0.047156	0.047829	0.985936	0.3242
MANUF	-0.145374	0.037743	-3.851646	0.0001
HEAD	-0.617408	0.082983	-7.440204	0.0000
TRUNK	-0.056082	0.052936	-1.059429	0.2895
UPEXTR	0.006121	0.036836	0.166170	0.8680
NECK	0.315990	0.121809	2.594149	0.0095
OCCDIS	0.380311	0.174518	2.179214	0.0294
HOSP	1.129203	0.036907	30.59574	0.0000

R-squared	0.183459	Mean dependent var	1.273806
Adjusted R-squared	0.181470	S.D. dependent var	1.275441
S.E. of regression	1.153925	Akaike info criterion	3.126830
Sum squared resid	7103.785	Schwarz criterion	3.144064
Log likelihood	-8348.706	Hannan-Quinn criter.	3.132850
F-statistic	92.20465	Durbin-Watson stat	1.939602
Prob(F-statistic)	0.000000		

Natural experiment

A **natural experiment** occurs when some exogenous event – often a change in government policy – changes the environment in which individuals, families, firms, or cities operate.

Control group is not affected by the policy change. **Treatment group** is thought to be affected by the policy change.

Call *C* the control group and *T* the treatment group, letting *dT* equal unity for those in the treatment group *T*, and zero otherwise. Then, letting *d2* denote a dummy variable for the second (post–policy change) time period, the equation of interest is

$$y = \beta_0 + \delta_0 d2 + \beta_1 dT + \delta_1 d2 * dT + other\ factors$$

δ_1 measures the effect of the policy. Without other factors in the regression, δ_1 will be the difference-in-differences estimator.

Illustration of the difference-in-differences estimator

$$y = \beta_0 + \delta_0 d2 + \beta_1 dT + \delta_1 d2 * dT + other\ factors$$

Difference-in-differences estimator (*without other factors in the regression*):

$$\hat{\delta}_1 = (\bar{y}_{2,T} - \bar{y}_{2,C}) - (\bar{y}_{1,T} - \bar{y}_{1,C})$$

	Before	After	After – Before
Control	β_0	$\beta_0 + \delta_0$	δ_0
Treatment	$\beta_0 + \beta_1$	$\beta_0 + \delta_0 + \beta_1 + \delta_1$	$\delta_0 + \delta_1$
Treatment – Control	β_1	$\beta_1 + \delta_1$	δ_1

Summary

Under reasonable assumptions, including the proxy variable in an OLS regression eliminates, or at least reduces, the omitted variables bias.

The method of instrumental variables is a different way to estimate the parameters in a linear model consistently when one or more explanatory variables are endogenous. The method of two stage least squares is used routinely in the empirical sciences.

Independent cross sections arise when different random samples are obtained in different time periods (usually years). OLS and the usual inference procedures are available, including corrections for heteroskedasticity. Because of the time series dimension, we often allow different time intercepts. We might also interact time dummies with certain key variables to see how they have changed over time. This is especially important in the policy evaluation literature for natural experiments.